

Mukkara Rithvik Reddy

✉ rithvikreddymukkara@gmail.com

☎ +91 88848 16582

🌐 [linkedin.com/in/rithvik-reddy-911b6b279](https://www.linkedin.com/in/rithvik-reddy-911b6b279)



github.com/RithvikReddy0-0

EDUCATION

Amrita Vishwa Vidyapeetham

B.Tech in Computer Science and Engineering

Bengaluru, India

Aug 2023 – Present

- **CGPA: 8.24/10** | Relevant Coursework: Machine Learning, Data Structures & Algorithms, Database Systems, Cloud Computing

CERTIFICATIONS

AWS Certified Cloud Practitioner Amazon Web Services | Validation: [d3aef95861bc4a029a6698e4f3474b64](https://aws.amazon.com/certification/verify/d3aef95861bc4a029a6698e4f3474b64) Issued May 2026 · Valid through May 2029

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, HTML/CSS

AI / ML / NLP: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, LangChain, SpaCy, NLTK, Gensim, OpenCV, Pandas, NumPy, Matplotlib, Seaborn

Cloud & Backend: AWS (Cloud Practitioner), FastAPI, Node.js, REST APIs, Vector Databases, Embeddings

Dev Tools: Git, LaTeX, Tailwind CSS, Bootstrap, BioPython, BLAST, JSON

PROJECTS

LegalEase | RAG Framework for Indian Legal Documents | Python, LangChain, LLMs, FastAPI, Vector DBs

Sep – Nov 2025

- Architected an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline enabling conversational querying of Indian legal and tax documents with citation-grounded responses.
- Engineered document chunking, vector embeddings, and semantic search to surface contextually relevant legal clauses; reduced LLM hallucinations by anchoring generation strictly to retrieved sources.
- Exposed inference via a **FastAPI** REST service; designed for scalable deployment on cloud infrastructure.

K-mer Genome ML Pipeline | Bioengineered Pathogen Detection | Python, Scikit-learn, Random Forest, BioPython

Sep – Nov 2025

- Built an alignment-free ML pipeline using **4-mer feature engineering** for plasmid sequence classification, processing and curating **113,000+ genomic sequences** with BLAST-based taxonomic validation.
- Achieved **98.98% classification accuracy** and **ROC-AUC of 0.9994** for engineered vs. natural plasmid detection; work submitted to peer-reviewed conference.

Uncertainty-Aware Crop Yield Prediction | Python, PyTorch, CNN-LSTM, Deep Ensembles

Sep – Dec 2025

- Designed a **CNN-LSTM multimodal fusion architecture** integrating satellite imagery, climate time-series, and agricultural yield datasets for spatiotemporal prediction.
- Applied **deep ensemble-based uncertainty estimation** to improve model robustness and generalization; conference submission planned.

InBetween | Geolocation-Based Social Meetup Platform | JavaScript, Tailwind CSS, SQL

Jun 2024 – Jan 2025

- Built and launched a geolocation-based web platform for peer discovery; scaled to **10,000+ visits**, validating product-market fit with 98% mobile device compatibility.

JIPMER Hallowen Festival Website | JavaScript, CSS, Tailwind, SQL

2024

- Delivered the official event website drawing **5,000+ unique visitors in 72 hours**; achieved **92/100 Google Lighthouse scores** across SEO, accessibility, and performance metrics.

RESEARCH & PUBLICATIONS

K-mer-Driven Genome-Informed Machine Intelligence for Detecting Bioengineered Pathogens — Published in *ICKECS 2026 Proceedings, SJCI Institute of Technology*, Apr 2026

Lexical Semantic Analysis in Gendered Bias: Designing Fairness Metrics for NLP Systems — Under Review, *Q2 Journal*

LEADERSHIP & AWARDS

Chairperson — IEEE Acrom Club (Robotics & Automation Society)

Dec 2024 – Present

Amrita Vishwa Vidyapeetham

Bengaluru

- Lead chapter operations, technical workshops, and inter-collegiate robotics initiatives; recognized with the **Best IEEE Student Branch Chapter Award (Dec 2025)**.

National Robotics Competition - 5th Place (NRC 2018) · 4th Place (NRC 2019)

2018 & 2019

Technoxian / IIT Partners